MIL-R-46332A(MU)
10 March 1965
SUPERSEDING
MIL-R-46332(ORD)
28 February 1962

MILITARY SPECIFICATION

REPRODUCER, SIGNAL DATA: AN/GSQ-64 (READER, PUNCHED TAPE)

1. SCOPE

1.1 This specification covers one type of signal data reproducer, punched tape reader. This device is powered by a 3 phase (4 wire) 400 cycle-per-second, alternating current 208 volt source. It will provide for rapid fill of an electrical memory from 0.6875 inch to 1.0 inch wide, fan-folded (8.5 inch folds), punched tape. It is referred to herein as the reproducer.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-T-60164	Tape, Fan Folded
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel; Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army Munitions Command

A8575680	Reproducer, Signal Data: AN/GSQ-64

FSC 1220

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PACKAGING DATA SHEET

MIL-P-14232/A8575680 Packaging of Reproducer, Signal Data: AN/GSQ-64

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The reproducer shall be manufactured in accordance with Drawing A8575680 and drawing pertaining thereto.

3.2 General specification.- The reproducer shall meet the requirements of Specification MIL-F-13926, as specified below. Should any conflict exist between the general specification and this specification, this specification shall take precedence.

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship
(f) Soldering electrical connections

3.3 Environmental service conditions.

3.3.1 Vibration.- The reproducer shall not be damaged by the vibration specified below:

(a) A simple harmonic motion of 0.015 inch amplitude (0.03 inch total excursion).

(b) A uniformly varied frequency of from 10 to 55 cycles-per-second (cps) and return to 10 cps in 1 minute ± 15 seconds.

(c) A vibration in 3 mutually perpendicular axes.

3.3.2 Shock.- The reproducer shall not be damaged by shock impulses of 15 gravity units (g's) of 11 ± 1 millisecond duration applied in 3 mutually perpendicular directions.

3.3.3 High and low temperature storage.- The reproducer shall not be damaged by periods of storage in a thermally stabilized condition at the ambient temperatures of +160°F. and -80°F.

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3.3.4 High and low operational temperature.-The reproducer shall meet the requirements of 3.4 through 3.6 within ± 5°F of the ambient temperature specified. Where none is specified in the individual requirements of 3.3 and 3.4 through 3.6 the specified performance shall be at an ambient temperature of between 60°F and 90°F.

3.3.5 Humidity.-The reproducer shall operate in an atmosphere having a relative humidity (RH) of 90 to 95 percent. After storage in an atmosphere with a RH of 100 percent including condensation it shall, when thoroughly dried, meet the requirements of 3.4 through 3.6.

3.3.6 Fungus.-The reproducer shall meet the requirements of 3.4 through 3.6 after exposure to fungus in accordance with Specification MIL-F-13926.

3.4 Electrical.

3.4.1 Input potential.-The reproducer shall meet the requirements of 3.4 through 3.6 upon application of 100 to 140 alternating-current volts (vac), at a frequency of 400 ± 5 cps existing between line and neutral in all three phases.

3.4.2 Output potentials; reproducer power supplies.-The reproducer shall deliver between the minimum and maximum values of voltage specified in Table I under the dc load conditions of normal operation. The aforesaid voltages of designated polarity, shall exist between test point jack J10, pin 4 and the pin connection given in Table I at the ambient temperatures specified.

3.4.3 Output ripple; reproducer power supplies.-The peak to peak ripple in millivolts (mv), existing between pin 4 of pin jack J10 (reproducer power supply ground) and the pin designated for each supply in requirement 3.4.3.1 to 3.4.3.5 inclusive, shall not exceed the value specified therein, when the reproducer is energized (power-on-state) under normal dc load conditions.

3.4.3.1 Positive six-volt supply.-The ripple voltage present between reproducer test-point jack J10 (pin 5) and reproducer power supply ground shall not exceed 10.0 mv.

3.4.3.2 Negative twelve-volt supply.-The ripple voltage present between reproducer test-point jack J10 (pin 6) and reproducer power supply ground shall not exceed 20.0 mv.

3.4.3.3 Negative fifteen-volt supply.-The ripple voltage present between reproducer test-point jack J10 (pin 13) and reproducer power supply ground shall not exceed 20.0 mv.

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3.4.3.4 Negative six-volt supply.-The ripple voltage present between reproducer test-point-jack J10 (pin 14) and reproducer power supply ground shall not exceed 20.0 mv.

3.4.3.5 Negative twenty-volt supply.-The ripple voltage present between reproducer test-point-jack J10 (pin 10) and reproducer power supply ground shall not exceed 100.0 mv.

3.4.4 Homing.-The reproducer shall home, upon passage of a punched tape (containing a homing slot) through its photodiode head, at an ambient temperature of between 60°F and 90°F and at the ambients of -40°F and +125°F. The homing slot shall consist of 10 consecutive sprocket holes between which the tape material has been removed.

3.4.5 Power phase detector.-When energized, the reproducer shall operate upon reversal of any two of the three-phase leads and not alter functional operation of the unit.

3.5 Punched tape.-The reproducer shall meet the requirements of 3.4 through 3.6 upon passage of a fan-folded punched-data-tape (8.5 inch folds) through the reproducer photodiode-head. This tape shall meet the requirements of MIL-T-60164.

3.5.1 Tape reading rate.-The nominal punched tape reading rate shall be 52 inches per second (520 frames per second plus or minus 10 percent) at ambient temperatures of between 60°F and 90°F and at the ambients of -40°F and at +125°F.

3.5.2 The reproducer shall continuously feed a 1000 foot length of .6875 inch wide punched tape, as specified in 3.5, from a tape cartridge to a tape canister. Oscilloscope presentation of pulses generated by at least one character line and the strobe line shall indicate uninterrupted reproducer performance at an ambient temperature between 60°F and 90°F and at ambients of -40°F and +125°F. The tape shall properly fold into the canister and the canister spring shall defect at a rate of 6 ± 0.5 inch per pound. The tape loading level shall remain within the prescribed limit lines of the tape canister.

3.5.3 Punched tape alinement.-The reproducer shall aline and seat at all points in its drive assembly tape widths of 0.6875 and 1.00 inch punched tape at ambient temperatures of between 60°F and 90°F and ambients of -40°F and at +125°F. The tape shall be as specified in 3.5.

3.5.4 Diameter and spacing of punched holes.-The diameter of a punched character hole shall be 0.072 + 0.001 or -0.002 inch. The diameter of a punched strobe hole shall be 0.046 + 0.001 or -0.002 inch. The spacing between adjoining punched holes on a single line shall be 0.1 ± 0.003 inch.

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TABLE I - ELECTRICAL VOLTAGES

Output Level	Test Point Jack Pin		Output DC Voltage at Plus 77°F		at Minus 40°F		at Plus 125°F	
	(Number)	(Polarity)	(Min)	(Max)	(Min)	(Max)	(Min)	(Max)
1	6	Negative	11.5	13.5	11.3	13.3	11.4	13.2
2	13	Negative	14.1	15.6	14.0	15.5	14.2	15.6
3	14	Negative	5.6	6.2	5.6	6.2	5.6	6.2
4	10	Negative	18.3	20.3	17.8	19.8	18.5	20.5
5	5	Positive	5.6	6.2	5.4	6.1	5.6	6.2

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3.6 Operational. - The requirements of 3.6.1 through 3.6.12 shall be met after compliance with condition (a) and (b) below.

(a) Channel bias level: A negative minimum bias 1.1 volt respective to ground shall exist between ground and the terminal board connector pin listed in 3.6(c) and within the range of the associated potentiometer. This bias voltage shall exist after setting in motion a tape (having only a sprocket line punched) and after R9 is adjusted to produce, between ground and pin 9 of J11, a positive width pulse which shall be 35% of the repetition rate.

(b) Functional pulse presentation. - After setting in motion a nine line tape (having eight channel lines and one sprocket line punched) the pulse in Figure 1 shall be obtained between the terminal board connector pin listed in 3.6(c) and ground within the range of the associated potentiometer.

(c)

Terminal Board Connector Pin	Associated Potentiometer
TB4-20	R1
TB4-18	R2
TB4-16	R3
TB4-14	R4
TB4-11	R5
TB4-9	R6
TB4-7	R7
TB4-5	R8

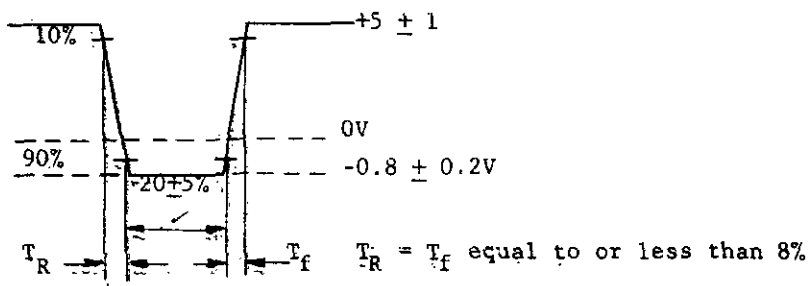


FIGURE 1. INPUT FUNCTIONAL PULSE

NOTE: The character pulse specified in 3.6.1 shall be a positive going voltage. The strobe pulse specified in 3.6.2 shall be a negative going voltage.

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3.6.1 Character pulses.-With the Tape Advance Switch in the OFF position, energizing the reproducer and electrically connecting pin W of Output Connector PB to pin 13 of J10 shall advance tape. The reproducer shall simultaneously deliver an oscilloscope presentation of the output pulse shown in Figure 2. Nominal pulse duration (t2) shall be 20 ± 5 percent of the entire 1.92 millisecond cycle (t1 + t2) shown in Figure 2. Under the above conditions the peak pulse amplitude shall be between the minimum and maximum values specified in Table II for the appropriate ambient temperature. This pulse shall be obtained across a 5,600 ohm resistive load existing between the reproducer output connector pin listed in Table II and power supply ground, pin N, at a point within the range of rotation of the respective potentiometer. Under all the conditions above, in addition to electrically connecting together reproducer output connector pins, A and B, there shall be obtained across a 150 ohm resistive load (when substituted for the 5,600 ohm resistive load) the voltage pulse shown in Figure 2. The peak amplitude of the positive going pulse shall be between -2.2 and -4.7 volts at the ambient temperatures specified in Table II. Pulse duration shall be as previously specified in this requirement.

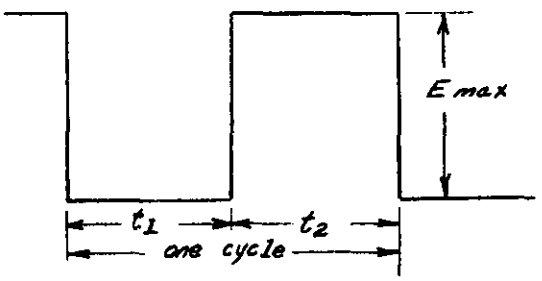


FIGURE 2. OUTPUT FUNCTIONAL PULSE

3.6.2 Strobe pulses.-With the tape advance switch in OFF position, energizing the reproducer and electrically connecting pin W of output connector PB to pin 13 of J10 shall advance the tape. The reproducer shall simultaneously deliver an oscilloscope presentation of the strobe (clock) pulse shown in Figure 3. Nominal pulse duration (t1) shall be 35 +15 - 0% of the entire 1.92 millisecond cycle (t1 + t2) shown in Figure 3. Under above conditions the peak pulse amplitude shall be between the maximum and minimum values specified in Table II for the appropriate ambient temperature. This pulse shall be obtained across a 5,600 ohm resistive load, existing between reproducer output connector pin S and power supply ground, pin N, and at a point within the range of rotation of potentiometer R9. Under all conditions above in addition to electrically connecting together reproducer output connector pins A and B, there shall be obtained across a 150 ohm resistive load (substituted for the 5,600 ohm resistive load) a negative going dc pulse. The peak amplitude of this pulse shall be between -14.0 and -16.0 volts at the ambient temperatures in Table II. Pulse duration shall be as previously specified in this requirement.

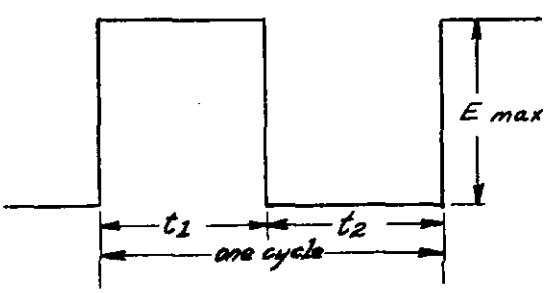


FIGURE 3. OUTPUT STROBE PULSE

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TABLE II. CHARACTER AND STROBE PULSES

Reproducer Output Connector Pin	Applicable Potentiometer	NEGATIVE VOLTAGE PULSE AMPLITUDE at Plus 77°F		at Minus 40°F		at Plus 125°F	
		(Min)	(Max)	(Min)	(Max)	(Min)	(Max)
D adjust	R_1	7.8	9.2	7.7	9.05	7.75	8.95
E adjust	R_2						
F adjust	R_3						
G adjust	R_4						
H adjust	R_5						
J adjust	R_6						
K adjust	R_7						
L adjust	R_8						
S adjust	R_9	14.1	15.6	14.0	15.5	14.2	15.6

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3.6.3 Functional TR output. - With the tape advance switch in OFF position (tape stopped), energizing the reproducer and electrically connecting pin W of output connector PB to pin 13 of J10 shall produce between 0 and -0.5 of a dc volt across pin M of output connector PB and power supply ground (pin N) at the ambient temperatures in Table I. When the tape advance switch is momentarily placed in ON position (tape running) the dc voltages shall be as specified in Level I of Table I at the ambient temperatures indicated.

3.6.4 Functional Rho output. - With the tape advance switch in OFF position and the reproducer energized, momentarily switching the tape advance switch to ON position shall set the tape in motion. Electrically connecting reproducer output connector, pin X, to pin 13 of J10 shall halt the tape.

3.6.5 Functional Rgo output. - With the tape advance swtich in OFF position and the reproducer energized, connecting pin W of plug PB to pin 13 of J10 shall cause the reproducer to produce output pulses on an oscilloscope at all temperatures between -40°F degrees and +125°F.

3.6.6 Functional GP rc' output. - When the reproducer is energized and the RUN-HALT switch is placed in HALT position, the negative voltage between reproducer output connector, pin d, and power supply ground pin N shall be as specified in Level 2 of Table I. There shall also exist between reproducer output connector, pin e, and power supply ground, pin N, a positive voltage equal to that specified in Level 5 of Table I. Voltages at each level shall be the value specified for each indicated ambient temperature.

3.6.7 Functional FB pr output. - When the reproducer is energized and the FADAC Start switch is held in START position, there shall appear between the reproducer output connector PB, pin f, and power supply ground, pin N, the Level 2 specified in Table I, for each indicated ambient temperature.

3.6.8 Functional CTTI output. - When the reproducer is energized and the Field Data Teletype switch is placed in TELETYPE position, the negative voltage between reproducer output connector, pin V, and power supply ground pin N shall be as specified in Level 3 of Table I for each indicated ambient temperature. The above voltages shall be obtained when pin C of output connector PB is electrically connected to pin 14 of J10.

3.6.9 Fill and verify indicator light function. - With the source of plus 145.0 vdc applied through a 100,000 ohm resistance to reproducer output connector, pin b, and the negative side of said source connected to the reproducer output connector pin h, there shall be illumination of the verify indicator light (neon light NE 12). When the negative side of above source is removed from reproducer output connector, pin h, and applied to reproducer output connector, pin g, there shall be illumination of the FILL indicator light (neon light NE 11).

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3.6.10 Auxiliary memory switch function. - Between reproducer output connector, pin T, and output connector pins Y, Z and a in sequence there shall exist less than one ohm of electrical resistance when the Auxiliary Memory switch is in OFF position. When the above switch is in ON position, there shall exist an electrical resistance in excess of 20 megohms.

3.6.11 Brake timing. - After insertion into the reproducer, an endless loop of tape with all strobe holes punched, plus one character hole in channel one, and the tape reader operating by energizing the reproducer, the tape shall proceed from OFF the read head when the Tape Advance switch is momentarily positioned from OFF to ON position then back to the OFF position. Connecting reproducer output connector, pin D, to pin X shall now halt the tape reader. The tape shall now be carefully held in the position assumed when the above stoppage occurred and upon moving away the head to expose the tape, examination shall indicate that no more than three strobe holes have passed beyond the character hole.

3.6.12 Photolamp potential. - The potential existing across the reproducer photolamps shall be 16.0 Vdc ± 5 percent at an ambient temperature of between 60°F and 90°F and at the ambient temperatures of -40°F and +125°F.

3.7 Pre-delivery. - Prior to shipment, and after all the previous requirements have been met, the reproducer shall again meet the requirements of 3.6a and 3.6b.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of product. - Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examination and tests. - Except where otherwise specified herein, components and sub-assemblies shall be inspected in accordance with the "Inspection provisions for components and sub-assemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to

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Section 3 herein shall be performed in accordance with Tables III, IV, & V and the test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.2.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.2.4 Functional tests. - The requirements and tests in Table I shall be inspected on a 100% basis. All examinations and tests shall conducted at the standard ambient temperature (60° to 90°F).

TABLE III

(100 percent inspection)

Characteristic	Requirements	Test Procedures
101. Input potential	3.4.1	4.5.2.1
102. Output potentials - power supplies	3.4.2	4.5.2.2
103. Output ripple: +6 volt supply	3.4.3, 3.4.3.1	4.5.2.3
104. Output ripple: -12 volt supply	3.4.3, 3.4.3.2	4.5.2.3
105. Output ripple: -15 volt supply	3.4.3, 3.4.3.3	4.5.2.3
106. Output ripple: -6 volt supply	3.4.3, 3.4.3.4	4.5.2.3
107. Output ripple: -20 volt supply	3.4.3, 3.4.3.5	4.5.2.3
108. Homing	3.4.4	4.5.2.4
109. Power phase detector	3.4.5	4.5.2.5
110. Punched tape	3.5	4.5.3
111. Tape reading rate	3.5.1	4.5.3.1
112. Tape feed	3.5.2	4.5.3.2
113. Punched tape alignment	3.5.3	4.5.3.3
114. Diameter and spacing of punched holes	3.5.4	4.5.3.4
115. Channel bias level	3.6(a), 3.7	4.5.4.1
116. Functional pulse presentation	3.6(b), 3.7	4.5.4.2
117. Character pulses	3.6.1	4.5.4.3
118. Strobe pulses	3.6.2	4.5.4.4
119. Functional TR output	3.6.3	4.5.4.5
120. Functional Rho output	3.6.4	4.5.4.6
121. Functional Rgo output	3.6.5	4.5.4.7
122. Functional GPrc' output	3.6.6	4.5.4.8
123. Functional FBpr output	3.6.7	4.5.4.9
124. Functional CTT1 output	3.6.8	4.5.4.10
125. Fill and verify indicator light function	3.6.9	4.5.4.11
126. Auxiliary memory switch function	3.6.10	4.5.4.12
127. Brake timing	3.6.11	4.5.4.13
128. Photolamp potential	3.6.12	4.5.4.14
129. Pre-delivery	3.7, 3.6(a) & (b)	4.5.5, 4.5.4.1, 4.5.4.2

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4.3 Standard sampling.

4.3.1 General. - One reproducer shall be selected at random by a Government representative as a standard sample from each 100 produced. The sample shall meet the requirements and tests in Table IV.

TABLE IV

Characteristic	Requirements	Test Procedures
301. Fabrication	3.1	Applicable drawings - visual
302. General specifications	3.2	MIL-F-13926 - visual

4.3.2 Environmental. - Three units shall be selected at random by a Government representative as a standard sample from each 50 produced or from each month's production, whichever occurs first. The samples shall have met the requirements and tests in Table III and shall then meet the requirements and tests in Table V.

TABLE V

Characteristic	Requirements	Test Procedures
303. Vibration	3.3.1	4.5.1.1
304. Shock	3.3.2	4.5.1.2
305. Humidity-operating and non-operating	3.3.5	4.5.1.3
306. Temperature (operating and storage extremes)	3.3.3, 3.3.4	4.5.1.4

4.3.3 Failure sample. - Should any one item of a standard sampling fail to meet the specified test requirements acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (See 4.2.3).

4.4 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design. - The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, demensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the

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test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.1.1 Test equipment. - In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5:

Item Number	Description
1. Multimeter	0 to 300 volts dc; 0 to 300 ma; 0-300 volts ac plus or minus 5 percent on any full scale reading
2. Oscilloscope	Dual Channel; 0.02 micro-second rise time; band width dc to 15 megocycles; sweep range: 1 micro-second to 10 milliseconds; internal and external triggering
3. Power Supply	150 volts dc; 10 milliamperes plus or minus 5 percent
4. Power Supply	208 volts; 400 cycles per second plus or minus 5 percent; 3-phase; 4 wire; 180 voltamps at 0.8 power factor
5. Resistors	5600 ohm plus or minus 5 percent; 150 ohms plus or minus 5 percent; 100K ohms plus or minus 10 percent
6. Tape I	Short endless, with sprocket and all (8) channels punched
Tape II	Short Endless with sprocket and homing slot punched
Tape III	1000 feet; in canister; one inch; all channels punched
Tape IV	Short endless, with all sprocket and one character hole punched
Tape V	Short endless tape with all sprocket holes punched
Tape VI	Same as Tape III except .6875 inch wide

4.4.1.1.1 Tape fan folded. - Fan folded tape to be used for tests herein shall be in conformance with MIL-T-60164.

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4.5 Test methods and procedures

4.5.1 Environmental.

4.5.1.1 Vibration. - The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.3.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.3.1 with the reproducer rigidly mounted in its normal operating position. Subsequent to the vibration test, the unit shall show no damage or defect resulting from the test when examined by visual and tactile means. The unit shall then be inspected as specified in 4.5.1.2.

4.5.1.2 Shock. - This test shall be conducted in accordance with the conditions of 3.3.2 and Procedure II of the shock test defined in MIL-F-13926 except that impact shocks shall not exceed 5G's. The reproducer shall be subsequently inspected for physical damage or defects in compliance with 3.3.2. The unit shall then be inspected as specified in 4.5.1.3.

4.5.1.3 Humidity. - The reproducer shall be tested in accordance with the conditions specified in 3.3.5. Distilled or demineralized water having a pH value between 6.5 and 7.5 at 77°F shall be used to obtain the required humidity. The reproducer shall be in operation during the humidity testing with a relative humidity of 90 to 95 percent for a period of not less than one hour. It shall then be subjected to the storage period of 4 hours with a relative humidity of 100% including condensation. After the 4 hour period the reproducer shall be thoroughly dired and examined for defects or damage. The reproducer shall then be subjected to the test specified in 4.5.2 to 4.5.4.1.4 inclusive to determine conformance with the requirements referenced therein.

4.5.1.4 Temperature (operating and storage). - After being subjected to all environmental tests specified above, the reproducer shall be subjected to Test Procedure I specified in MIL-F-13926, except the temperatures specified in 3.3.4 shall apply. While at the operating temperatures of 3.3.4 the unit shall be tested in accordance with 4.5.2 to 4.5.4.14 inclusive to determine compliance with 3.3.4. The temperature shall then be returned to standard temperature (60° to 90°F) and the unit shall be tested in accordance with 4.5.2 to 4.5.4.14 inclusive to determine compliance with 3.3.3.

4.5.1.5 Fungus. - To determine compliance with 3.3.5, one of the first 5 reproducers produced shall he submitted for testing in accordance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.

4.5.2 Electrical.

4.5.2.1 Input potential. - Remove the front panel assembly. Connect power supply (Item 4 of 4.4.1.1) to the reproducer connector PA. Measure the voltages between line and neutral, in all three phases, with a multimeter (Item 1) to determine compliance with the conditions specified in 3.4.1.

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4.5.2.2 Output potentials; reproducer power supplies. - Set the reproducer power switch and signal switch to the "ON" position. With multimeter (item 1) measure the voltages at the pins specified in Table 1 and test point jack pin number 4. The voltages shall be within the limits specified in Table 1 when tested at each of the temperatures specified therein to determine compliance with 3.4.2.

4.5.2.3 Output ripple; reproducer power supplies. - Energize the reproducer as specified in 4.5.2.2. Connect the oscilloscope (Item 2) to the pins as specified in Table VI. The ripple voltages shall not exceed the limits shown in Table VI and as specified in 3.4.3.1 to 3.4.3.5 inclusive.

TABLE VI

Reproducer power supplies	between pins of J10	peak-to-peak ripple
positive six-volt	5 and 4	10.0 millivolts
negative twelve-volt	6 and 4	20.0 millivolts
negative fifteen volt	13 and 4	20.0 millivolts
negative six-volt	14 and 4	20.0 millivolts
negative twenty-volt	10 and 4	100.0 millivolts

4.5.2.4 Homing. - Set power and signal switches to the "OFF position. Set FILL-VERIFY switch in Fill position. Load Tape II (Item 6 of 4.4.2.3) into the read head. Start the tape by placing Tape Advance switch to the "ON" position. The tape advance shall be stopped when the homing slot is positioned in the photo-diode read head. Set the Fill-Verify switch to the Verify position, observing that the tape advances continuously. After meeting this test at the standard ambient temperature (60°F to 90°F) in compliance with 3.4.4 the test shall be repeated at minus 40°F and plus 125°F.

4.5.2.5 Power phase detector. - Halt tape reader and remove tape from read head. Reverse any 2 of the 3 phase input power leads (Item 4) to determine compliance with 3.4.5. Start the unit and determine is the capstan continues to rotate in counterclockwise direction when viewed from the front.

4.5.3 Punched tape. - Tape II (Item 6 of 4.4.2.3) shall be visually inspected for any defects and shall conform to the requirements specified in 3.5. Tape III shall be used in the following tests to determine conformance with punched tape requirements referenced therein.

4.5.3.1 Tape reading rate. - Place Tape III into read head. Turn TAPE ADVANCE switch to the "ON" position, advancing the tape for three seconds as measured with a stop-watch. Push "stop switch," stopping the tape advance. The measured linear advance shall be 156 inches ± 10 percent to determine compliance with 3.5.1. After meeting this test at the standard ambient temperature (60° to 90°F) the test shall be repeated at minus 40°F and plus 125°F.

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4.5.3.2 Tape feed. - Place tape VI into read head. Connect channel 1 of the oscilloscope (Item 2) between pin S and ground, and channel 2 between any of the following pins of connector PB: D, E, F, G, H, and ground. Advance the tape for the time specified in 3.5.2. The oscilloscope presentation of the character pulses of channel 2 shall be positive going, and of a duration between 15-25 percent of the repetition rate. The strobe signal shall be negative going, and between 35-50 percent of the repetition rate in duration. This test shall be performed with the power supply (Item 4) adjusted to 100, 120 and 140 Vac ± 5 percent. After meeting this test at the standard ambient temperature the test shall be repeated at minus 40°F and plus 125°F.

4.5.3.3 Punched tape alinement. - The punched tapes of widths specified in 3.5.3 shall be placed and seated in the drive assembly. The setting of each type shall be visually observed for compliance with the requirements of 3.5.3. After meeting this test at the standard ambient temperature the test shall be repeated at minus 40°F and plus 125°F.

4.5.3.4 Diameter and spacing of punched holes. - A tool maker's microscope shall be utilized to measure the diameters and spaces of the punched holes to determine compliance with the requirements specified in 3.5.4.

4.5.4 Operational.

4.5.4.1 Channel bias level. - Insert Tape V (Item 6 of 4.4.1.1) into the read head. Connect oscilloscope between one of the terminal board connector pins and power ground. Set power and signal switches to the "ON" position. The amplitude of voltage as measured on the screen shall be as specified in 3.6(a) when each potentiometer is adjusted. Repeat the test at each of connector pins with adjustments of related potentiometers specified. Disconnect the switches and remove tape. Repeat test prior to delivery as specified in 3.7.

4.5.4.2 Functional pulse. - Insert Tape I into read head and reset the switches back to the "ON" position. The oscilloscope screen shall measure the pulse shown in Figure I. Repeat test at each connector pin with adjustment of related potentiometers specified in 3.6(b). Place switches backto the "OFF" position. Repeat test prior to delivery as specified in 3.7.

4.5.4.3 Character pulses. - An oscilloscope (Item 2) shall be connected between the respective output connector pins and power supply ground pin specified in 3.6.1. The amplitude of the pulses shall be as shown in Table II at each of the pins as measured on the oscilloscope screen. The output connector pins A and B then shall be shorted . The amplitude of the pulses shall be between-2.2 to +4.7 volts at each of the respective pins as measured on the oscilloscope screen. Pulse duration (t_1) shall measure 80 ± 5% of the entire cycle on the screen when the applicable potentiometers listed in the Table II are adjusted by means of a screw driver for optimum character pulse presentation. After meeting this test at the standard ambient temperature the test shall be repeated at minus 40°F and plus 125°F to determine compliance with the values specified in Table II for these temperatures.

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4.5.4.4 Strobe pulse. - The oscilloscope (Item 2) shall be connected between reproducer output connector (PB) pin S and ground pin N. The amplitude of the pulse shall measure between minus 14.0 and minus 16.0 volts. Pulse duration (t1) shall measure a nominal 35 percent of the entire cycle as shown on the oscilloscope screen. After meeting this test as the standard ambient temperature the test shall be repeated at minus 40°F and plus 125°F to determine compliance with 3.6.2 and the values specified in Table II for these temperatures.

4.5.4.5 Functional Tr output. - The Tape Advance Switch shall be placed in the "OFF" position. Output connector pins W and N shall be shorted. A multimeter, Item 1, shall be placed between pin M and pin N. The measured voltage shall be between 0 and minus 0.5 volt dc. The Tape Advance Switch shall then be momentarily placed in the "ON" position. The measured voltage shall be as indicated in 3.6.3 After meeting this test at the standard ambient temperature the test shall be repeated at minus 40°F and plus 125°F.

4.5.4.6 Functional Rho output. - Activate the reproducer momentarily turn Tape Advance switch to the "ON" position. The output connector pin X shall then be connected to test point jack pin 13, halting the tape reader operation. After meeting this test of the standard ambient temperature in compliance with 3.6.4, the test shall be repeated at minus 40°F and plus 125°F.

4.5.4.7 Functional Rgo output. - Activate the reproducer then connect pin W and test point pin 13 to start tape. After meeting this test at the standard ambient temperature in compliance with 3.6.5, the test shall be repeated at minus 40°F and plus 125°F.

4.5.4.8 Functional GP rc' output. - The reproducer shall be energized. The "Run-Halt" switch shall be placed in the "Halt" position. A multimeter (Item 1) shall be placed between pin d and pin N, and the voltage measured shall be as indicated in 3.6.6. The measured voltage between pin e and pin N shall be positive as indicated in 3.6.6. After meeting this test of the standard ambient temperature, the test shall be repeated at minus 40°F and plus 125°F.

4.5.4.9 Functional FBpr output. - The reproducer shall be energized. Hold FADAC start switch in "Start" position. The voltage measured with multimeter between pin f and pin N shall be as indicated in 3.6.7. After meeting this test at the standard ambient temperature the test shall be repeated at minus 40°F and plus 125°F.

4.5.4.10 Functional CTTI output. - The reproducer shall be energized and the "Field Data Teletype" switch placed in Teletype position. Connect pin C of PB to pin 14 of J10. The measured voltage across pin V of PB and ground pin N, shall be as specified in 3.6.8. After meeting this test at the standard ambient temperature the test shall be repeated at minus 40°F and plus 125°F.

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4.5.4.11 Fill and Verify indicator light function. - The reproducer input power shall be disconnected. A 100,000 ohm resistor shall be connected to pin B. Power supply, Item 3, shall connected in series with the load resistor to pin b, with the ground side to pin g. This will illuminate the "Fill" neon; with the ground side connected to pin h, the "verify" neon shall light. After meeting this test at the standard ambient temperature the test shall be repeated at minus 40°F and plus 150°F.

4.5.4.12 Auxiliary memory switch function. - All input power shall be disconnected. A miltimeter, Item 1, shall be connected between the pins specified in 3.6.10.. Place Auxiliary memory switch to the "OFF" position. Resistance measured across the specified pins shall read less than one ohm. Placing switch in the "ON" position, the meter reading shall indicate more than 20 megohms.

4.5.4.13 Brake timing. - Place Test Tape IV in the read head and momentarily place the Tape-Advance switch on the "ON" position to start tape operating. Connecting jumper from pin D to pin X shall halt the tape within three sprocket holes beyond the single character hole in channel one to determine compliance with 3.6.11.

4.5.4.14 Photolamp potential. - The potential between the photolamps shall be measured with the multimeter (Item 1). The voltage shall be as specified in 3.6.12. After meeting this test at the standard ambient temperature the test shall be repeated at minus 40°F and plus 125°F.

4.5.5 Pre-delivery tests. - Prior to shipment and subsequent to having passed all testing above, the reproducer shall again be tested in accordance with 4.5.4.1 and 4.5.4.2 to determine compliance with 3.7 and requirements referenced therein.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - The reproducer shall be packaged, packed and marked in accordance with Specification MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P8575680. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The reproducer is for use as input equipment for the rapid fill of an electrical memory of a digital computer such as the Gun Direction Computer M18, with a predetermined program from a punched taped. It may be also be used to provide a test signal as when used in conjunction with the Computer Logic Test Set, AN/GSM-70.

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6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of an applicable level of preservation, packaging and packing
(c) Applicable stock number
(d) Applicable packaging data sheet (see 5.1)
(e) A block of serial numbers to identify each reproducer

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Preparing activity:

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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